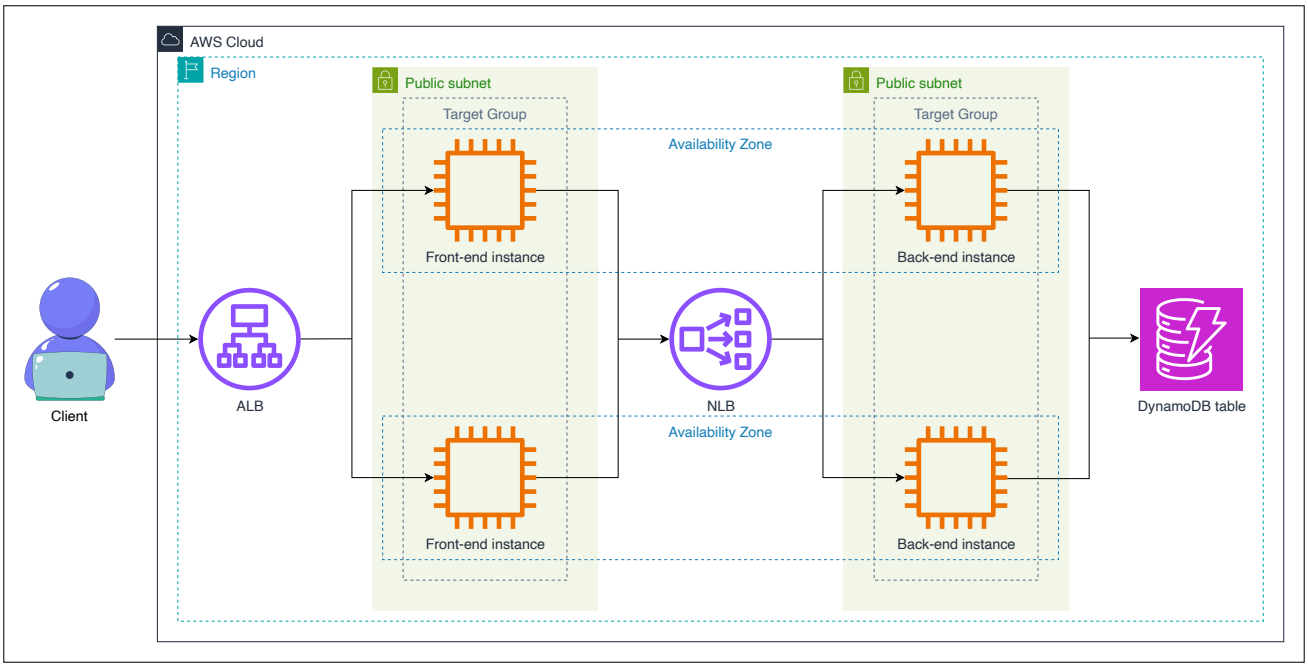
Overview of Elastic Load Balancing (ELB)

Explore Elastic Load Balancing to distribute application traffic efficiently across multiple AWS resources. Understand the differences between Application, Network, and Gateway Load Balancers and how to apply them for scalable, fault-tolerant architectures.

Imagine we're deploying a modern web application that needs to serve thousands—or even millions—of users worldwide. One of the first architectural problems we'll face is traffic distribution. That's where Elastic Load Balancing (ELB) steps in. **ELB** is an AWS-managed service that automatically distributes incoming application traffic across multiple targets like Amazon EC2 instances, containers, and IP addresses within one or more Availability Zones.

Scalability and availability are key in cloud-native applications. Load balancing ensures that no single component becomes a bottleneck or point of failure. Instead of relying on a single backend server, ELB routes traffic dynamically to healthy targets. This approach not only increases reliability but also enhances performance.

Ask



Managing ALBs using ELBs

ELB integrates deeply with AWS Availability Zones (AZs), allowing us to distribute application traffic across multiple isolated locations. This architecture minimizes the risk of a single point of failure.

Types of load balancers

ELB comes in three flavors; each load balancer type serves a distinct purpose, and selecting the right one often depends on the nature of our traffic and application requirements.

Application Load Balancer

Application Load Balancer (ALB) operates at the application layer (layer 7) of the OSI model. It intelligently routes requests based on URL path, hostname, query string, HTTP headers, and more. This makes it a perfect fit for modern web apps that rely on microservices or container-based architectures. For example, if we run an e-commerce app, ALB can route traffic to `/checkout` endpoints to one and `/products` to another, all on the same domain.

Ask

A common real-world use case can be routing user requests based on the URL path or host (e.g., sending `/login` traffic to an authentication service and `/shop` traffic to a catalog service).

Network Load Balancer

Network Load Balancer (NLB) works at layer 4 (transport layer) and is optimized for ultra-low latency and high throughput. It's capable of handling millions of requests per second while maintaining ultra-fast response times. NLB is often used for applications that require fast TCP or UDP traffic handling, such as gaming servers, real-time financial apps, or VoIP systems. It also supports static IP addresses and Elastic IPs, making it ideal for integrating with legacy systems or on-prem connections.

In the real world, an example could of TCP/UDP-based applications (e.g., multiplayer game servers using UDP for real-time gameplay).

Understanding the difference between Layer 4 and Layer 7 load balancers is critical when designing architectures:

Feature	Application Load Balancer (Layer 7)	Network Load Balancer (Layer 4)
Protocols	HTTP, HTTPS	TCP, UDP, TLS
Routing logic	URL path, headers, host	IP and port
Performance	Moderate	Ultra-low latency
Health checks	HTTP/HTTPS-based	TCP-based
SSL Termination	Yes	No (handled by targets)

Ask

Gateway Load Balancer

Gateway Load Balancer (GWLB) is a unique breed. It acts as a transparent bump-in-the-wire device, directing traffic to virtual appliances like firewalls, intrusion detection/prevention systems, or deep packet inspection tools. One of GWLB's strengths is traffic mirroring, where a copy of the network traffic is sent to the monitoring appliance without affecting the original traffic flow.

One particular application could be centralizing traffic inspection (e.g., routing VPC traffic through a central firewall).

> **Performance tip:** Choose ALB for smart and NLB for raw speed.

Real-world architecture scenarios

Let's examine a few real-world scenarios.

- **Scenario 1:** We deploy a web app with distinct microservices for user authentication, catalog browsing, and payments. This scenario benefits from fine-grained Layer 7 routing, which allows targeted traffic delivery to individual services.
- **Scenario 2:** For a trading app requiring sub-millisecond latency, NLB can distribute TCP traffic across high-performance EC2 instances. Its low-latency performance and static IP support make it ideal for real-time financial workloads.
- **Scenario 3:** We're integrating a third-party firewall into our cloud environment. GWLB allows seamless traffic redirection through the firewall without altering the network architecture. This approach simplifies the deployment of virtual appliances at scale while maintaining network transparency.

Ask

Conclusion

Elastic Load Balancing (ELB) is a cornerstone of scalable, fault-tolerant